

Odd coefficients in quasifibonacci products

A signed-cancellation proof of the statement of
Stanley's Conjecture 6.3

Uniform in the recurrence order, with arbitrary
superincreasing seeds and exact algorithms

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Main result. Let $k \geq 2$. For any positive integer seeds $w_1, \dots, w_k$ with $w_i > \sum_{j < i} w_j$, continue by $w_n = w_{n-1} + \dots + w_{n-k}$. If $h_k(n)$ counts the odd coefficients of $\prod_{i=1}^n (1 + x^{w_i})$, then

$$\sum_{n \geq 0} h_k(n) z^n = \frac{1 + 2z^k}{1 - 2z + 2z^k - 2z^{k+1}}.$$

The answer depends on k and n , not on the particular seeds.

Status. The report supplies a self-contained mathematical proof and reproducible exact checks. The displayed identity is explicitly conjectured in the cited version of Stanley's paper. Historical priority is not established by this report, and the proof has not been independently refereed or formalized.

Abstract

We study the parity of coefficients of finite products whose exponents satisfy a k -term Fibonacci recurrence. The target is the rational generating function stated in Stanley's Conjecture 6.3. We prove that identity for every $k \geq 2$ and, more generally, for every quasifibonacci sequence with superincreasing initial seeds. The key construction changes selected plus signs in the product to minus signs: a periodic sign block of length $k + 1$ with product -1 makes every coefficient belong to $\{-1, 0, 1\}$. Since signs disappear modulo two, a parity count becomes a squared-coefficient norm. An exact overlap identity then gives a recurrence of order $k + 1$. We also obtain a positive recurrence, a colored composition interpretation, minimality of the recurrence order, exponential asymptotics, and a coefficient-parity algorithm requiring only a single branch at each step. All principal identities are proved independently of the computational checks included in the archive.

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1 The selected problem and the scope of the result

1.1 A documented conjecture

Fix $k \geq 2$. In Stanley's notation, the k -bonacci numbers have initial values

$$F_1^{(k)} = \cdots = F_k^{(k)} = 1, \quad F_{r+1}^{(k)} = F_r^{(k)} + \cdots + F_{r-k+1}^{(k)} \quad (r \geq k).$$

Set

$$I_n^{(k)}(x) = \prod_{i=1}^n (1 + x^{F_{i+k-1}^{(k)}}). \quad (1.1)$$

Let $h_k(n)$ be the number of its coefficients that are odd, with $h_k(0) = 1$. The statement labeled Conjecture 6.3 in arXiv:2101.02131v3 is precisely the identity on the title page [1, §6, p. 24]. The paper was published in *European Journal of Combinatorics* in 2024. We use the conjecture numbering of the cited arXiv version throughout.

This is a finite-product problem: n is the number of factors, rather than an upper bound on the exponent being counted. Zeros inside the polynomial's coefficient range are harmless because they are not odd. Coefficients beyond the degree are also not counted.

1.2 What is proved, and what is not claimed

The main theorem proves the stated identity uniformly in k . The proof does not infer a recurrence from finitely many terms. Instead, it establishes an identity for every coefficient in the overlap of two translated products and then sums it exactly. It also shows that the same formula works for arbitrary superincreasing seeds, not just the seeds in (1.1).

There is important prior work to acknowledge. Zhao proved flatness of finite signed Fibonacci products when $k = 2$ [3]. Diao introduced the quasifibonacci seed conditions used below and proved flatness of the infinite all-minus product for even recurrence order [2, Definition 1.1 and Theorem 1.1]. The argument here uses a varying, periodic signing, works for both even and odd orders, concerns every finite product, and gives its exact support count. These distinctions matter: simply replacing every plus sign by a minus sign does *not* work for odd k .

The literature check for this report examined the primary sources listed in the bibliography, including the available Stanley version and related later work. No later proof of this particular all- k parity formula was located. That is a bounded literature-search statement, not a certification that no such proof exists. The mathematical claim made here is the validity of the proof below; a claim of first publication would require a separate review.

The result does not settle Stanley's general coefficient-residue conjecture for every modulus, nor the higher-moment identities in the same paper.

2 A stronger theorem for arbitrary seeds

Definition 2.1. A *quasifibonacci sequence of order k* is a sequence of positive integers $w_1, w_2, \dots$ satisfying

$$w_i > S_{i-1} := \sum_{r=1}^{i-1} w_r \quad (1 \leq i \leq k), \quad (2.1)$$

$$w_n = \sum_{r=n-k}^{n-1} w_r \quad (n \geq k+1). \quad (2.2)$$

We put $S_0 = 0$ and $S_n = \sum_{r=1}^n w_r$. The term agrees with Diao's definition, with "level" replaced by "order."

The seed condition is often called *superincreasing*: each seed is larger than all earlier seeds combined. For instance, $(1, 3, 5)$ and $(2, 7, 15)$ are valid order-three seeds; $(1, 2, 3)$ is not.

Definition 2.2. A sign sequence $\varepsilon_1, \varepsilon_2, \dots \in \{-1, 1\}$ is *coherent* if

$$\varepsilon_n = - \prod_{r=n-k}^{n-1} \varepsilon_r \quad (n \geq k+1). \quad (2.3)$$

Its first k signs can be chosen arbitrarily. A convenient canonical choice is

$$\varepsilon_i = \begin{cases} -1, & k+1 \mid i, \\ 1, & k+1 \nmid i. \end{cases} \quad (2.4)$$

Define the ordinary and signed products by

$$I_n(x) = \prod_{i=1}^n (1 + x^{w_i}), \quad A_n(x) = \prod_{i=1}^n (1 + \varepsilon_i x^{w_i}) = \sum_{m \in \mathbb{Z}} a_n(m) x^m.$$

The coefficient function $a_n(m)$ is zero outside $0 \leq m \leq S_n$.

Theorem 2.3 (Uniform parity and signed flatness). *For every sequence in Definition 2.1, every coherent signing, and every $n \geq 0$, the following conclusions hold.*

- (i) Every coefficient of A_n is $-1, 0$, or 1 .
- (ii) For each $m \in \mathbb{Z}$, the residue modulo two of $[x^m]I_n$ is $a_n(m)^2$.
- (iii) The number of odd coefficients is independent of the seeds and satisfies

$$h_k(n) = 2^n \quad (0 \leq n \leq k), \quad (2.5)$$

$$h_k(n) = 2h_k(n-1) - 2h_k(n-k) + 2h_k(n-k-1) \quad (n \geq k+1). \quad (2.6)$$

Consequently,

$$H_k(z) := \sum_{n \geq 0} h_k(n) z^n = \frac{1 + 2z^k}{1 - 2z + 2z^k - 2z^{k+1}}. \quad (2.7)$$

The next five sections prove the theorem. The proof has two distinct parts: first flatness, then an exact norm recurrence. The latter alone would not justify counting odd coefficients without the former.

3 Arithmetic of the exponent sequence

Lemma 3.1 (Separation). *The weights are strictly increasing. For every $j \geq 0$,*

$$w_{j+2} > S_j, \quad 2w_{j+1} > S_j. \quad (3.1)$$

Proof. The seeds are strictly increasing by (2.1). At the first recurrence step, $w_{k+1} = S_k > w_k$. Each subsequent weight is a sum of $k \geq 2$ positive weights including its immediate predecessor, so strict increase persists.

For $0 \leq j \leq k-2$, the first inequality follows directly from the seed condition. For $j = k-1$, we have $w_{k+1} = S_k > S_{k-1}$. If $j \geq k$, the recurrence includes the two preceding weights, so induction gives

$$w_{j+2} \geq w_{j+1} + w_j > S_{j-1} + w_j = S_j.$$

This proves the first inequality at all indices. For $j = 0$, the second inequality is immediate. For $j \geq 1$, use the first inequality one index lower together with strict increase:

$$S_j = S_{j-1} + w_j < w_{j+1} + w_{j+1}.$$

□

Lemma 3.2 (Exact overlap length). *For $n \geq k+1$, put $j = n - k - 1$. Then*

$$S_{n-1} = w_n + S_j. \quad (3.2)$$

Thus the coefficient ranges of $A_{n-1}(x)$ and $x^{w_n}A_{n-1}(x)$ overlap exactly in the interval $[w_n, w_n + S_j]$.

Proof. Split S_{n-1} after its first j summands. The remaining k summands have sum w_n by (2.2). The interval assertion follows because $\deg A_{n-1} = S_{n-1}$ and both endpoint coefficients are nonzero. Here “coefficient range” means the containing interval, not necessarily a support without internal gaps. □

Lemma 3.3 (The original seeds are admissible). *For $w_i = F_{i+k-1}^{(k)}$, the first k weights are*

$$w_1 = 1, \quad w_i = (k-1)2^{i-2} + 1 \quad (2 \leq i \leq k), \quad (3.3)$$

and satisfy $w_i - S_{i-1} = k - i + 1 > 0$ for $2 \leq i \leq k$.

Proof. The first new term after the k initial ones is k . During the subsequent initial steps, moving the recurrence window replaces one of those initial ones by the newest term, so the new value is twice its predecessor minus one. This proves (3.3) by induction. Summing that expression gives

$$S_{i-1} = (k-1)(2^{i-2} - 1) + (i-1),$$

which yields the claimed difference. The later weights satisfy (2.2) by their definition. □

4 Coherent signs and reflection

Lemma 4.1 (Periodic description of all coherent signings). *A signing is coherent if and only if it is periodic with period dividing $k+1$ and every consecutive block of length $k+1$ has product -1 . There are 2^k such signings.*

Proof. Equation (2.3) says that each length- $(k + 1)$ window has product -1 . Dividing the products for two consecutive windows cancels the k common signs and gives $\varepsilon_{r+k+1} = \varepsilon_r$. Conversely, a periodic sign block with product -1 has that product in every window, and hence satisfies (2.3). Its first k signs are free and its final sign is their negative product. The canonical choice (2.4) is the block of k plus signs followed by one minus sign. $\square$

Put $E_n = \prod_{i=1}^n \varepsilon_i$, including $E_0 = 1$.

Lemma 4.2 (Reflection). *For every integer m ,*

$$a_n(S_n - m) = E_n a_n(m). \quad (4.1)$$

If $n \geq k + 1$ and $j = n - k - 1$, then

$$E_{n-1} E_j = -\varepsilon_n. \quad (4.2)$$

Proof. For an individual factor, $x^{w_i}(1 + \varepsilon_i x^{-w_i}) = \varepsilon_i(1 + \varepsilon_i x^{w_i})$. Multiplication gives $x^{S_n} A_n(x^{-1}) = E_n A_n(x)$ and proves (4.1), including the zero coefficients outside the range. Since $E_j^2 = 1$, the product $E_{n-1} E_j$ consists precisely of the k signs from $j + 1 = n - k$ through $n - 1$. Its value is $-\varepsilon_n$. $\square$

5 The pointwise cancellation identity

This section contains the decisive calculation. It is an identity over the integers, before reducing anything modulo two.

Proposition 5.1 (Two disjoint tails). *Let $n \geq k + 1$, $j = n - k - 1$, and $v = w_{j+1}$. For $0 \leq u \leq S_j$,*

$$\boxed{a_n(w_n + u) = \varepsilon_n \varepsilon_{j+1} (a_j(u - v) - a_j(u + v))}. \quad (5.1)$$

The two terms on the right cannot both be nonzero.

Proof. By Lemma 3.1, the factors of index $j + 2$ or higher have all their nonconstant terms in degrees larger than S_j . Therefore, on the interval $0 \leq u \leq S_j$,

$$a_{n-1}(u) = a_j(u) + \varepsilon_{j+1} a_j(u - v). \quad (5.2)$$

There is no convergence or truncation approximation in this equation: later factors simply cannot contribute to such a low exponent.

Reflect the high end of A_{n-1} using $S_{n-1} = w_n + S_j$ and (5.2). We obtain

$$\begin{aligned} a_{n-1}(w_n + u) &= E_{n-1} a_{n-1}(S_j - u) \\ &= E_{n-1} (a_j(S_j - u) + \varepsilon_{j+1} a_j(S_j - u - v)) \\ &= E_{n-1} E_j (a_j(u) + \varepsilon_{j+1} a_j(u + v)) \\ &= -\varepsilon_n (a_j(u) + \varepsilon_{j+1} a_j(u + v)). \end{aligned} \quad (5.3)$$

Now use $A_n = A_{n-1} + \varepsilon_n x^{w_n} A_{n-1}$ and add (5.3) to ε_n times (5.2). The two copies of $a_j(u)$ cancel exactly, leaving (5.1).

For the first tail to be nonzero, $u \geq v$ is necessary. For the second, $u \leq S_j - v$ is necessary. These conditions cannot hold simultaneously, because $2v > S_j$. More explicitly, the possible tail intervals are

$$[0, S_j - v] \quad \text{and} \quad [v, S_j],$$

with an interval understood to be empty if its endpoints are reversed. They are disjoint. $\square$

Remark 5.2. At the very first overlap, $n = k + 1$ and $j = 0$. Both tails are empty: $a_0(-w_1) = a_0(w_1) = 0$. Thus the only overlapping coefficient, at exponent $w_{k+1} = S_k$, cancels. This is an important endpoint check on the indexing.

6 Proof of flatness and the parity lift

Proof of Theorem 2.3(i)–(ii). For $0 \leq n \leq k$, the seed condition ensures that two distinct subsets of $\{w_1, \dots, w_n\}$ cannot have the same sum. Indeed, at their largest index of disagreement, the larger seed exceeds the sum of all earlier seeds. Every nonzero coefficient of A_n is therefore a single product of signs and is 1 or -1 .

Proceed by strong induction on n . For $n \geq k + 1$, outside $[w_n, S_{n-1}]$ at most one of the two terms A_{n-1} and $\varepsilon_n x^{w_n} A_{n-1}$ contributes. The coefficient there is zero or, up to sign, an earlier coefficient. Inside the overlap, Proposition 5.1 expresses it as the signed difference of two coefficients of A_j , where $j < n$, with at most one nonzero term. The induction hypothesis consequently proves flatness everywhere.

Modulo two, $\varepsilon_i = 1$, so $I_n \equiv A_n$ in $\mathbb{F}_2[x]$. A flat integer coefficient is odd exactly when its square is one. Therefore

$$([x^m]I_n \bmod 2) = a_n(m)^2 \in \{0, 1\}. \quad (6.1)$$

This proves both assertions. $\square$

The result can also be read as a signed enumeration statement. Among all subsets having a specified weight, split them according to the product of their assigned signs. The sizes of the two classes differ by at most one. This signing is not necessarily parity of the subset's cardinality. For odd k , that distinction is exactly what makes the theorem possible.

Corollary 6.1 (An infinite-product extension). *For any admissible seeds and coherent signing, the formal infinite product $\prod_{i \geq 1} (1 + \varepsilon_i x^{w_i})$ has all coefficients in $\{-1, 0, 1\}$.*

Proof. The weights tend to infinity. For each fixed exponent, all sufficiently late factors contribute only their constant term, so the coefficient stabilizes to one of the finite-product coefficients already proved flat. $\square$

For even k , choosing all initial signs negative makes every sign negative; this recovers the setting of Diao's infinite-product theorem. For odd k , a periodic coherent signing replaces that all-minus choice.

7 The exact norm recurrence and generating function

Define

$$q_n = \|A_n\|_2^2 := \sum_{m \in \mathbb{Z}} a_n(m)^2, \quad B_n = \sum_{m \in \mathbb{Z}} a_{n-1}(w_n + m) a_{n-1}(m).$$

All sums here and below are finite. Expanding the square in $A_n = A_{n-1} + \varepsilon_n x^{w_n} A_{n-1}$ gives

$$q_n = 2q_{n-1} + 2\varepsilon_n B_n. \quad (7.1)$$

Flatness and (6.1) imply $q_n = h_k(n)$, but we first derive the norm recurrence directly.

Lemma 7.1 (The overlap inner product). *For $n \geq k + 1$ and $j = n - k - 1$,*

$$\varepsilon_n B_n = q_j - q_{j+1}. \quad (7.2)$$

Proof. Only $0 \leq u \leq S_j$ can contribute to B_n . Substitute (5.2) and (5.3), abbreviating $v = w_{j+1}$ and $e = \varepsilon_{j+1}$:

$$\varepsilon_n B_n = - \sum_{u=0}^{S_j} (a_j(u) + e a_j(u+v)) (a_j(u) + e a_j(u-v)).$$

The product $a_j(u+v)a_j(u-v)$ vanishes for every u , by the same separation $2v > S_j$ used in Proposition 5.1. The two mixed sums are equal after a change of index. If

$$T = \sum_{u \in \mathbb{Z}} a_j(u) a_j(u-v),$$

then the preceding display becomes $\varepsilon_n B_n = -q_j - 2eT$. On the other hand, multiplication of A_j by $1 + ex^v$ yields $q_{j+1} = 2q_j + 2eT$. Eliminating T proves $\varepsilon_n B_n = q_j - q_{j+1}$. $\square$

Proof of Theorem 2.3(iii). The superincreasing seed block gives $q_n = 2^n$ for $0 \leq n \leq k$. Substitution of (7.2) into (7.1) gives

$$q_n = 2q_{n-1} - 2q_{n-k} + 2q_{n-k-1} \quad (n \geq k+1).$$

Since $q_n = h_k(n)$, this proves (2.5)–(2.6). Neither the coefficients of this recurrence nor its initial values depend on the seeds or on the first k signs.

Finally multiply $H_k(z)$ by $D_k(z) = 1 - 2z + 2z^k - 2z^{k+1}$. Its constant coefficient becomes one. Its coefficients of degrees $1, \dots, k-1$ vanish, its coefficient of degree k is two, and all later coefficients vanish by the recurrence. Thus $D_k H_k = 1 + 2z^k$, proving (2.7). $\square$

By Lemma 3.3, this is a proof of the statement of Stanley's Conjecture 6.3. It also proves the seed-independent extension in Theorem 2.3.

8 Examples, checks, and the role of the signs

8.1 The first genuinely overlapping order-three example

The standard order-three weights are

$$1, 3, 5, 9, 17, 31, 57, 105, 193, \dots$$

The canonical coherent signs repeat $+, +, +, -$. For the first four factors,

$$\begin{aligned} A_4(x) &= (1+x)(1+x^3)(1+x^5)(1-x^9) \\ &= 1 + x + x^3 + x^4 + x^5 + x^6 + x^8 \\ &\quad - x^{10} - x^{12} - x^{13} - x^{14} - x^{15} - x^{17} - x^{18}. \end{aligned}$$

There are fourteen nonzero coefficients. The ordinary all-plus product has a coefficient two at x^9 and fourteen odd coefficients elsewhere. The recurrence gives the same answer: $h_3(4) = 2 \cdot 8 - 2 \cdot 2 + 2 \cdot 1 = 14$.

A later example displays the disjoint-tail mechanism rather than just a single cancellation. At $n = 8$, we have $j = 4$, $v = 17$, and $S_j = 18$. The overlap is $105 \leq m \leq 123$. Formula (5.1) allows nonzero coefficients there only at $m = 105, 106, 122, 123$; each is -1 . The fifteen intermediate positions cancel.

8.2 Why the naive all-minus choice fails

For the same weights,

$$[x^9](1-x)(1-x^3)(1-x^5)(1-x^9) = -2.$$

The contributions come from the singleton $\{9\}$ and the triple $\{1, 3, 5\}$; both have negative cardinality sign. An odd-order recurrence replaces k selected parts by one part without changing cardinality parity. The coherent signing instead forces those two contributions to have opposite signs. This explains why the proof must change the signing, not merely reuse the all-minus Fibonacci product.

8.3 Initial counts and an OEIS connection

n	$h_2(n)$	$h_3(n)$	$h_4(n)$	$h_5(n)$	$h_6(n)$
0	1	1	1	1	1
1	2	2	2	2	2
2	4	4	4	4	4
3	6	8	8	8	8
4	8	14	16	16	16
5	12	24	30	32	32
6	20	40	56	62	64
7	32	68	104	120	126
8	48	116	192	232	248
9	72	200	356	448	488
10	112	344	660	864	960
11	176	592	1224	1668	1888
12	272	1016	2272	3220	3712

For $k = 2$, the sequence satisfies $h_2(n) = 2A104767(n)$ for $n \geq 1$, with the separate value $h_2(0) = 1$. The cited OEIS entry records the corresponding Fibonacci signed-product interpretation [4]. No new OEIS identifier is assigned or implied for the tables accompanying this report.

Remark 8.1 (The seed hypothesis cannot simply be omitted). For $k = 3$ and seeds $(1, 2, 3)$, the first three all-plus factors already have an even coefficient at exponent three and only six odd coefficients, not 2^3 . Likewise, duplicate order-two seeds $(1, 1)$ give $(1+x)^2$, with two odd coefficients instead of four. The theorem is not asserting the same initial conditions for arbitrary positive seed lists.

9 A positive recurrence and a finite counting formula

The minimal recurrence in (2.6) has a negative coefficient. A slightly longer one has only nonnegative coefficients and makes both positivity and asymptotic analysis more transparent.

Put

$$B_k(z) = 1 + z + \cdots + z^{k-1}, \quad L_k(z) = z + z^2 + \cdots + z^{k-1} + 2z^{2k}.$$

Direct multiplication gives

$$B_k(z)D_k(z) = 1 - L_k(z). \tag{9.1}$$

Indeed, $(1-z)B_k = 1 - z^k$, and the terms of degree k cancel in $B_k(1-2z) + 2z^k(1-z)B_k$.

Corollary 9.1 (Positive recurrence). *For every $n \geq 2k$,*

$$h_k(n) = \sum_{r=1}^{k-1} h_k(n-r) + 2h_k(n-2k). \quad (9.2)$$

Equivalently,

$$H_k(z) = \frac{(1 + 2z^k)B_k(z)}{1 - L_k(z)}. \quad (9.3)$$

Proof. Multiply numerator and denominator of (2.7) by B_k and use (9.1). The numerator has degree $2k - 1$, so extracting coefficients in degrees at least $2k$ proves (9.2). $\square$

Formula (9.3) is the generating function of the following simple objects. Choose a prefix of size $d + ek$, where $0 \leq d < k$ and $e \in \{0, 1\}$; the prefix has one color if $e = 0$ and two colors if $e = 1$. Follow it by an ordered sequence of parts of sizes $1, 2, \dots, k-1, 2k$, where the part $2k$ has two colors and the other parts each have one color. Total size is counted by z .

This is an exact equinumerosity obtained from the generating function. It is not claimed to be an explicit bijection with the odd coefficient positions of the original product.

In particular, an entirely finite, nonrecursive expression is

$$h_k(n) = \sum_{e=0}^1 \sum_{d=0}^{k-1} \sum_{\substack{m_1, \dots, m_{k-1}, b \geq 0 \\ \sum_{r=1}^{k-1} r m_r + 2kb = n - d - ek}} 2^{e+b} \frac{(m_1 + \dots + m_{k-1} + b)!}{m_1! \dots m_{k-1}! b!}. \quad (9.4)$$

An impossible constraint contributes zero. The multinomial coefficient orders the chosen parts; the power of two supplies their colors and, when applicable, the prefix color. This proves (9.4) directly from the description above.

Proposition 9.2 (Minimal recurrence order). *Over $\mathbb{Q}$, the smallest order of an eventual homogeneous constant-coefficient recurrence for $h_k(n)$ is $k + 1$.*

Proof. The denominator D_k has degree $k + 1$. It is relatively prime to $N_k(z) = 1 + 2z^k$, because

$$D_k(z) = (1 - 2z)N_k(z) + 2z^{k+1}.$$

A common root would have to be zero, but $N_k(0) = 1$. Thus the fraction in (2.7) is reduced. If a recurrence with denominator polynomial $Q(z)$ holds eventually, $Q(z)H_k(z)$ is a polynomial, so the reduced denominator D_k divides Q . Its degree is therefore a lower bound for the recurrence order. The recurrence already proved attains it. $\square$

10 Exponential asymptotics and density

10.1 The unique dominant root

Let r_k be the unique positive solution of $L_k(r_k) = 1$, and write $\lambda_k = r_k^{-1}$.

Theorem 10.1 (Asymptotic count). *For every fixed $k \geq 2$, there is a number $0 < \mu_k < \lambda_k$ such that*

$$h_k(n) = C_k \lambda_k^n + O(\mu_k^n), \quad C_k = \frac{(1 + 2r_k^k)B_k(r_k)}{r_k L'_k(r_k)} > 0. \quad (10.1)$$

Moreover $1 < \lambda_k < 2$, and λ_k is the unique real root in $(1, 2)$ of

$$\lambda^{k+1} - 2\lambda^k + 2\lambda - 2 = 0. \quad (10.2)$$

Proof. L_k is strictly increasing on the positive real axis, with $L_k(0) = 0$ and $L_k(1) > 1$. Thus $r_k \in (0, 1)$ exists uniquely and $L'_k(r_k) > 0$. Also

$$L_k(1/2) = 1 - 2^{1-k} + 2^{1-2k} < 1,$$

so $r_k > 1/2$.

For complex z with $|z| < r_k$, the triangle inequality gives $|L_k(z)| \leq L_k(|z|) < 1$. If $|z| = r_k$ and $L_k(z) = 1$, equality in the triangle inequality forces all the summands to have the same argument, and their sum is positive real. The term z occurs with positive coefficient, so z itself is positive real. Hence $z = r_k$. This proves that r_k is the only zero of $1 - L_k$ on its smallest root circle, and it is simple.

By (9.3), the singular part of H_k at r_k is $C_k/(1 - z/r_k)$, with C_k as displayed. Subtracting it leaves a rational function analytic in a disk of radius strictly larger than r_k . Cauchy's coefficient bound on a circle inside that disk gives the exponentially smaller error term in (10.1). Finally, (9.1) and $B_k(r_k) > 0$ give $D_k(r_k) = 0$; substituting $r_k = 1/\lambda_k$ yields (10.2). Uniqueness follows from the uniqueness of the positive root of $1 - L_k$ and the fact that B_k is positive on the positive axis. $\square$

k	λ_k	C_k	ρ_k
2	1.543689012692	1.473679689278	1.618033988750
3	1.716672749282	1.550023534317	1.839286755214
4	1.855679027977	1.362757096516	1.927561975483
5	1.930604988558	1.200842855191	1.965948236645
6	1.966580689288	1.109329020459	1.983582843424
7	1.983724513901	1.059879860829	1.991964196605
8	1.991997413559	1.032931667570	1.996031179735

The decimal values in the table were computed by bisection with 80-digit working precision. They illustrate the theorem; the root-location and dominance assertions were proved without numerical root finding.

10.2 Odd coefficients have zero density in the full range

Let $\rho_k \in (1, 2)$ satisfy $\sum_{i=1}^k \rho_k^{-i} = 1$, the growth equation of the exponent sequence. For every admissible seed list, $S_n = \Theta(\rho_k^n)$. One can see this without a spectral theorem: choose positive constants bounding w_i/ρ_k^i above and below for the first k indices, and propagate the bounds through (2.2). Summing preserves the same exponential order.

Put $s = \rho_k^{-1}$. Since the k terms $s, \dots, s^k$ sum to one, we have $s^k < 1/2$. Consequently

$$L_k(s) = 1 - s^k + 2s^{2k} < 1,$$

so $s < r_k$ and $\lambda_k < \rho_k$. It follows that

$$\frac{h_k(n)}{S_n + 1} = O((\lambda_k/\rho_k)^n) \longrightarrow 0. \quad (10.3)$$

The denominator counts every coefficient position from zero to the degree, including positions with coefficient zero. No assertion about density *among nonzero coefficients only* is implicit in (10.3).

10.3 Large-order expansion

There is also a useful asymptotic as $k \rightarrow \infty$:

$$\lambda_k = 2 - 2^{1-k} - (2k - 4)4^{-k} + O(k^2 8^{-k}). \quad (10.4)$$

Here is a justification that keeps the varying parameter under control. Write $t = 2^{-k}$ and $\delta = 2 - \lambda_k$. Equation (10.2) becomes

$$\delta = 2t(1 - \delta)(1 - \delta/2)^{-k}. \quad (10.5)$$

For all sufficiently large k , the equivalent function $f(\delta) = \delta(2 - \delta)^k - 2 + 2\delta$ is negative at zero and positive at $4t$, since $f(4t) = 4(1 - 2t)^k - 2 + 8t \rightarrow 2$. The uniqueness already proved places the relevant root in $(0, 4t)$. Thus $k\delta \rightarrow 0$, and Taylor expansion, uniformly on that interval, gives

$$(1 - \delta)(1 - \delta/2)^{-k} = 1 + \frac{k-2}{2}\delta + O(k^2\delta^2).$$

First (10.5) yields $\delta = 2t + O(kt^2)$; substituting this back gives $\delta = 2t + 2(k-2)t^2 + O(k^2t^3)$, proving (10.4). This expansion is separate from the fixed- k , large- n asymptotic in (10.1).

11 Exact algorithms

11.1 Counting without expanding a polynomial

The recurrence computes $h_k(0), \dots, h_k(N)$ in $O(N)$ integer arithmetic operations after the initial block is available. A streaming implementation needs only $k + 1$ past values. The included `count_prefix` routine returns the whole list, so it intentionally uses $O(N)$ integer storage instead.

For an isolated very large index, let

$$\chi_k(X) = X^{k+1} - 2X^k + 2X - 2.$$

Compute $X^N \bmod \chi_k(X)$ by binary powering. If the remainder is $\sum_{i=0}^k b_i X^i$, then

$$h_k(N) = \sum_{i=0}^k b_i 2^i. \quad (11.1)$$

To justify this, let E be the shift operator on the sequence. The proved recurrence says $\chi_k(E)h = 0$ beginning at index zero in the shift-operator form, so polynomial division applied to E^N gives (11.1). Naive polynomial multiplication and reduction use $O(k^2 \log(N+1))$ ring operations. The same code works modulo any positive integer; it does not assume the modulus is prime.

These are arithmetic-operation bounds, not bit-complexity bounds. In exact arithmetic, $h_k(N)$ itself has $\Theta(N)$ bits for fixed k , by Theorem 10.1. Writing it already takes linear time in its bit length.

11.2 A single coefficient's parity in linear many steps

The pointwise proof gives an efficient coefficient oracle. Precompute the weights, partial sums, and signs through index n . To evaluate $a_n(m)$, first return zero if $m \notin [0, S_n]$. For $n > k$:

- (1) If $m < w_n$, replace (n, m) by $(n-1, m)$.

- (2) If $m > S_{n-1}$, replace (n, m) by $(n - 1, m - w_n)$ and multiply the accumulated sign by ε_n .
- (3) Otherwise set $j = n - k - 1$, $u = m - w_n$, and $v = w_{j+1}$. Formula (5.1) gives at most one continuing branch: use $a_j(u - v)$ if $u \geq v$; use $-a_j(u + v)$ if $u + v \leq S_j$; otherwise return zero. In either continuing case multiply also by $\varepsilon_n \varepsilon_{j+1}$.

When the index is at most k , a greedy subset-sum test on the superincreasing seeds finishes the computation. At every step the index strictly decreases; there is no branching tree. Hence the query uses $O(n)$ arithmetic comparisons, additions, and sign changes after $O(n + k)$ -entry preprocessing (including validation of the seed block). The absolute value of the resulting signed coefficient is the requested parity of the all-plus coefficient.

For fixed k and seed bit length b , the stored exponents have $O(n + b)$ bits. The arithmetic-step description therefore does not assert constant-time operations on arbitrarily large exponents. Its principal advantage is avoiding storage proportional to the exponentially large degree.

11.3 Direct multiplication as an independent check

An integer bitset represents a polynomial over $\mathbb{F}_2$ by storing its coefficient of x^m in bit m . Multiplication by $1 + x^w$ is exactly

```
bits ^= bits << w
```

and the final odd-coefficient count is `bits.bit_count()`. This is an excellent small-instance oracle, but its memory requirement is proportional to the polynomial degree. The supplied implementation has an explicit degree cap to prevent accidental enormous allocations.

The signed sparse and dense checks perform integer polynomial multiplication, not just multiplication modulo two. They therefore test the stronger flatness claim and the sign-sensitive overlap identity as well.

12 Reproducibility and computational audit

The archive includes a dependency-free Python 3.10+ implementation. The recorded run used Python 3.13.5 and passed every assertion. The substantive checks were as follows:

Check	Recorded coverage
Direct bitset products	519 prefix comparisons for $2 \leq k \leq 25$, with an 8,000,000-degree cap; the maximum n depends on k .
Signed dense products	144 products: $2 \leq k \leq 10$, $1 \leq n \leq 16$; 19,338 coefficient-oracle comparisons.
Every initial signing	All 2^k choices for $2 \leq k \leq 6$, two seed families, $n = 12$: 248 cases.
Random seed/sign cases	96 superincreasing seed lists and initial signings, each at $n = 12$, with a fixed random seed.
Pointwise overlap formula	5,821 exact coefficient identities for $2 \leq k \leq 6$, through $n = 14$.
Positive recurrence	28,101 comparisons for $2 \leq k \leq 30$, through $n = 1000$.
Fast versus linear counts	Eight indices per k for $2 \leq k \leq 30$, in exact arithmetic and modulo 1,000,000,007.

Coverage is intentionally reported with its limitations. In particular, the degree cap means that the largest tested orders in the bitset sweep reach only the nonoverlapping initial block. They are not evidence of long-run behavior at those orders. Nontrivial overlap checks occur at the smaller orders, and the proof, rather than the sweep, establishes all orders and all indices.

The generated data include counts through $n = 200$ for $2 \leq k \leq 20$, 80-digit working-precision growth constants, complete decimal values of $h_k(10000)$ at six orders, and large-index modular values. Some compact checkpoints are shown below.

k	Decimal digits of $h_k(10000)$	$h_k(10^{12}) \bmod 1000000007$
2	1886	338509426
3	2348	699562826
4	2686	896800000
5	2858	178164254
10	3007	587625022
20	3011	367480980

The complete large integers are in `data/large_values.json`; each was obtained both by the linear recurrence and by polynomial powering. Those two algorithms share the proved recurrence, so their agreement checks implementation, not the combinatorial theorem. The direct polynomial checks provide the independent small-instance comparison.

12.1 Commands

From the archive's top-level directory:

```
python code/verify.py
python code/kbonacci_parity.py 3 20 --prefix
python code/kbonacci_parity.py 3 10000
python code/kbonacci_parity.py 3 1000000000000000 --modulus 10000000007
python code/kbonacci_parity.py 3 8 --coefficient 105
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The last coefficient query returns signed coefficient -1 and parity one. The numeric algorithms require no computer algebra system. The PDF build uses standard TeX packages; the source and generated table fragments are included. No downloaded font files or third-party paper copies are bundled.

13 Further questions and boundaries of the method

The proof exposes several natural next questions without resolving them. For moduli greater than two, changing a factor's sign no longer leaves its residue unchanged, so the parity lift is unavailable in this form. A finite collection of root-of-unity-weighted products might replace it, but flatness over $\{-1, 0, 1\}$ by itself supplies no such result.

The positive composition model provides an enumerative explanation for $h_k(n)$ after the rational function has been proved. An explicit bijection between its colored objects and odd coefficient positions would give a different kind of explanation. None is asserted here.

It is also reasonable to ask which weaker seed conditions preserve the same count or admit a related rational function. The proof uses the seed condition twice: to obtain the initial values

2^n , and to establish the separation inequalities that make the tails disjoint. Changing only the initial values without checking that separation would not justify reusing the argument.

Finally, coherent signs were shown to be sufficient and completely parameterized within the coherence rule. We did not classify *all* possible sign sequences whose finite products happen to be flat. Such a necessity theorem is a separate question.

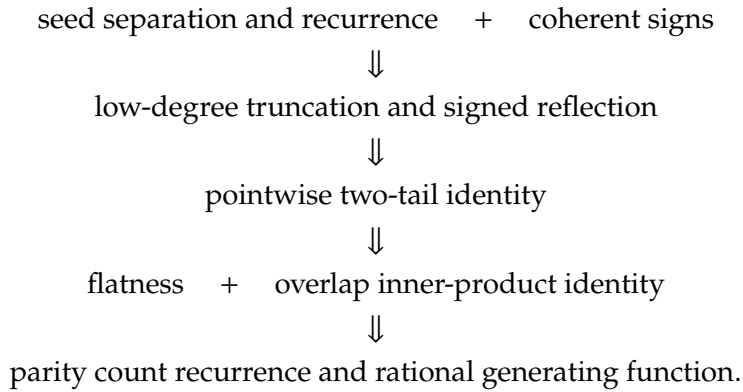
14 Conclusion

The selected conjectural identity follows from a simple structural fact: with an appropriate periodic signing, the large central overlap cancels, leaving two tails that are too far apart to interact. This proves flatness, turns parity into a squared norm, and reduces the norm update to a seed-independent recurrence.

The resulting theorem covers every recurrence order $k \geq 2$, every admissible seed list, every coherent signing, and every finite number of factors. Its consequences include an exact rational generating function, a minimal-order recurrence, an alternative positive recurrence, a nonrecursive finite formula, precise exponential growth, and practical exact algorithms. These conclusions are supported by the proofs above; the attached computations are a reproducible audit rather than a substitute for proof. Independent mathematical review and a fuller priority check remain appropriate before presenting the result as a new publication.

A A compact proof dependency map

The logical order is



In particular, neither an assumed rational generating function nor any unproved description of a word-rewriting monoid is used in the main argument. All boundary cases are covered by the convention that coefficient functions vanish outside their finite ranges.

B Archive contents and source-status notes

`article.tex` and `article.pdf` contain this report. `code/kbonacci_parity.py` provides the public algorithms; `code/verify.py` performs the recorded exact checks and regenerates the numerical data; `code/test_api.py` tests boundary cases and input validation. The data directory contains count tables, growth constants, large values, the verification log, and TeX table fragments. `README.md`, `RESEARCH_STATUS.md`, and the `Makefile` describe use, provenance, and rebuilding.

The main source is the versioned Stanley preprint, not an inferred statement from a search snippet. Its conjecture page was checked directly. The older even-order flatness result was also checked in Diao's primary preprint. The $k = 2$ finite-product result was checked in Zhao's paper. Related symbolic-generation work and its later solved challenge were reviewed to distinguish this problem from already resolved neighboring tasks [5, 6]. The present theorem does not claim to resolve that separate challenge or to replace those authors' algorithms.

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- [6] Jinlong Tang and Guoce Xin, *Meeting a challenge raised by Ekhad and Zeilberger related to Stern's triangle*, arXiv:2506.13375, 2025. [arXiv:2506.13375](#).